Jagath Vallabhai Missula

Senior Technical Lead

A technically astute and competent with more than 13.5 years of captivating experience, viz., 2.5 years in team leading, 13.5 years in software and 8 years in hardware development. Seeking career advancement with esteemed organization to leverage in engineering projects to drive innovation, optimize process, and deliver exception results to contribute the growth and success of a forward-thinking organization

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SKILLS

Product & application | Team manage & Lead | Quality Management & Safety Compliance | Operation & Control | Software- Architecture & Design | Simulation | MATLAB (M script, MIL, Simulink, HDL coder, Simscape, Simulink Extras) | PSCAD | PSpice | LTspice | Hardware | KEIL | Code composer studio (CCS) | Altium PCB | Power electronics | Distributed generation | Solar | Wind | Automotive | Electric Vehicles (EVs) | Electrical drives | Power quality | Assembly level language, FORTRAN, and C | MS office | MS visio, Entrprise Architecture, IOdraw | GIT, GERRIT, SVN, JIRA

KEY STRENGTHS

Domain Expertise

- Profound knowledge on defining the product specification, design algorithms, hardware boards.
- Proven R&D experience in developing the new concepts, including the strong literature, development of hardware topologies, software architecture and control algorithms.
- Worked in several phases of hardware development life cycle including system requirements, component selection, high-level and low-level design, development, system integration, functional testing and documentation.
- Worked in all the phases of software development cycle including software requirements, high-level and low-level design, development, integration, unit and functional testing and documentation.
- Proven technical and analytical skills in troubleshooting circuits and finding root cause of failures.

Engineering Leadership Expertise

- Proficient in overseeing the entire engineering scope from customer requirements, utilizing various functional resources for effective planning, monitoring and successful delivery. Adept at leading deliverables while maintaining high quality and meeting deadlines.

Risk Mitigation

- Demonstrated ability to mitigate engineering risks through cross-functional collaboration, and implementing timely risk mitigation strategies to ensure project success. Skilled in understanding the requirements, providing regular project updates, identifying potential challenges, and proactively taking actions to meet requirements.

EDUCATION

Qualified	Board/University	Institute	Percentage/Grade	Year
Ph. D. [Power Electronics, Solar and Wind Power Systems]	------	Indian Institute of Technology (IIT) Guwahati	---	2014–2022
M. Tech. [Electrical drives]	------	Maulana Azad National Institute of Technology (NIT), Bhopal	8.71 GGPA (Excellent)	2008–2010
B. Tech. [Electrical & electronics]	Jawaharlal Nehru Technological University (JNTU) Hyderabad	Raghu Engineering College, Visakhapatnam	68.65 % (First Class)	2004–2008
Intermediate [M. P. C.]	Board of Intermediate Education, Andhra Pradesh	Sri Chaitanya Junior College, Visakhapatnam	84.10 % (Distinction)	2002–2004
S. S. C.	Board of Secondary Education, Andhra Pradesh	Ramakrishna Residential Public School, Visakhapatnam	74.83 % (First Class)	2001–2002

ACHIEVEMENTS

- Project Prolificacy: Proven track record of delivering complex research projects.
- Cost Savings and Quality Enhancements: Improved production quality through the innovative 4Q approach to address manufacturing-related quality issues.

- Cost Saving Initiatives: Spear headed cost-saving initiatives, resulting in achieving reduction in cost, significantly contributing to operational efficiency.

AWARDS & HONORS

- Recognized in the AESTHETIC category for the *Robotic Car* project organized by IEEE Students branch in Robocamp'06 program held at International Institute of Information Technology (IIIT) Hyderabad.
- Volunteer service award for *18th National Power Systems Conference* (NPSC-2014) held at IIT Guwahati.

ACCOMPLISHMENTS

- Stood first for two semesters and overall in the M. Tech. branch.
- Secured full marks in Mathematics course in the second year of Intermediate.
- Secured full marks in Electronic devices & circuits course in the B. Tech. programme.
- Secured AIR **2054** in *GATE- 2008* examination and **95.15** Percentile in the *GATE- 2011* examination.

WORK EXPERIENCE - PROJECTS

Technical Lead @ Tech Mahindra Pvt. Ltd.

Sep., 2024 – Present

- Bipolar HVDC MIL test including test cases scripting in MATLAB-2024b SIMULINK.
- Cross collaboration with the project development and product improvement teams to timely update the test system.
- Training the new team members and supporting the core team members on the test system.

Senior Technical Lead/Product Owner @ TATA ELXSI Ltd.

Jul., 2023 – Aug., 2024

Safety implementation of traction power inverter for EVs

- Requirements (Technical and FuSa-ISO 26262 standards) drafting, high-level and low-level designing and MATLAB based ASW implementation.

Implementation of a grid connected converter with SRF-dq controller for UPF operation using MATLAB.

Software Defined Vehicles (SDV): Various EV features integration with vehicle and cloud

Leading and overseeing project activities with a strict focus on maintaining engineering quality, adhering to cost constraints, and meeting project timelines. Developing and executing project engineering plans, meticulously outlining objectives, strategies and resource allocation.

- Representing engineering within project core team and managing effective communication with internal stakeholders to ensure alignment and collaboration.
- Spearhead engineering risk mitigation efforts and devising action plans to safeguard project interest, viz., planning and scheduling the necessary reviews, diligently tracking and closing open action items within specified timelines, working on tactical issues for setting priorities, monitoring progress of programs, developing corrective actions when problems arose, schedule estimates, manpower estimates, and resource allocation.
- Playing pivot role in enhancing project cost-efficiency by spearheading cost on track program, i.e., development of common platform requirement strategies for senior management and improved recurring cost management.
- Project organization effectively using JIRA and GERRIT tools, viz., creating, ordering and managing product backlogs including establishment of group's goals & engineering objectives for design, development, testing and validation to accomplish major milestones, maintain project momentum and ensure their timely resolution.
- Providing management with MoMs, updates, data driven project performance analyses and comprehensive reports to enable informed decision-making.
- Driving and motivating the cross-functional teams to deliver the product as per the agreed-to requirements and cross functional management of other team members.
- Functionally managing the engineering core team, viz., setting KPIs for team, conducting one-on-one meetings to give/seek feedback with team members, monitoring performance, conducting appraisals to ensure their commitments align with the project objectives.
- Analysis of technical problems and not only assisted in providing recommendations, but also involved in providing high-level solutions.
- Proposal and high-level design of software architecture for EV features (Range polygon, Range estimation and BMS) and their deployment in the vehicle and as a digital twin in the cloud. These features communicate with various modules as per the architecture and publish the results in cluster/HMI using various interfaces.
- Owning project and technical review documents, viz., software requirements, low-level and high-level design, testing, verification, validation, etc., for supporting program manager as per NPD and NPI processes.

Technical Lead @ Embitel Technologies India Pvt. Ltd.

Dec., 2022 – Jul., 2023

PMSM and BLDC motor control and power electronic converter solutions using MATLAB

- Presenting engineering perspectives and providing a comprehensive overview of design process, issues, risks and solutions.

- Representing engineering within project core team and managing effective communication with internal stakeholders to ensure alignment and collaboration.
- Functionally managing the engineering core team and ensuring their commitments align with project objectives.
- Reviewing the customer project proposals, procurement, system and software requirement documents.
- Implementing risk management processes, mitigating project risks and ensuring schedule adherence.
- Validating the customer requirements, align seamlessly with offered engineering solutions through the submission of technical proposals.
- Ensuring timely input/output and design reviews for internal team, guaranteeing that these processes meet high quality standards and contribute to project success.

Advanced Engineer @ ePropelled Systems Pvt. Ltd.

Aug. – Dec., 2022

Three-phase PMSM drive for three wheeler electric vehicle application

- Sending RFQs and POs, obtaining quotations and PIs from the suppliers for various electronic components/equipment.
- Selection of components such as Hall Effect sensor, hex buffer IC, throttles, shielded/unshielded header connector, multi-core shielded/unshielded interfacing cables.
- Design of OPAMP, transistor and optocoupler based analog and digital signal conditioning circuits and their implementation using LTspice simulation and on hardware to produce the signals, viz., throttle, forward, brake and reverse to interface with the Digital Signal Controller (DSC).
- Carried out Altium PCB software based schematic and layout design for Rotary position sensor (Encoder).
- Managing the electrical production BoM's for encoder, DSC and inverter boards. Updating & Releasing BoM's.
- Power budget and cost budget preparation for the encoder, DSC and inverter boards.
- Design, hardware implementation, bulk testing and validation of encoder circuit.
- LTspice based simulation of three-phase inverter for sizing the DC bus capacitor by analyzing its ripple current and ripple voltage.
- Testing of PMSM drive with varying throttle input under no-load condition.
- Supported operation with production boards, resolving design and production issues.

Ph. D. thesis/Research scholar @ IIT Guwahati

Jul., 2014 – Aug., 2022

Average Modeling and DC-Link Capacitor Voltage Regulation of SRF-dq *Controlled Single-Phase Active Neutral Point Clamped Inverter (ANPCI) for Solar and Wind Power Applications*

- ANPCI HW program to bring the product from program kick-off to low-volume production in less than a year. Product is also achieved one of the highest levels of quality for all of the new design technology & topology.
- Streamlining the research HW project delivery plan covering all activities needed, viz., planning, execution, verification, and validation to ensure a fully verified and validated product including functions and deliverables.
- Carried out functional block diagrams, concept and design through full production.
- Carried out selection of power supply, DSO equipment, and components, viz., DSC, power semiconductors, gate drive ICs, voltage and current sensors, inductors, capacitors, connectors, low power transformers, etc., with power consumption estimation.
- Carried out Altium PCB software based schematic and layout design for ANPCI and dc-dc chopper circuits including analog/digital circuits, power converter switches (IGBTs), gate drive ICs, voltage and current sensor circuits, and their interfaces, etc.
- Carried out design to cost methodology activity considering reliability and performance under thermal and electrical stress, viz., cost and BoM components reduction, and obsolescence management.
- Defined PCB, hardware build-up with PCBA prototype, debugging, qualification and released products to the manufacturing.
- Proposed a capacitor voltage balancing strategy for 5L- ANPCI. Various capacitor voltage balancing strategies are customized and implemented for 5L-ANPCI. Further, their performance is compared with the proposed strategy.
- Various PWM techniques are customized for ANPCI and their performance is compared using PSCAD/EMTDC.
- ANPCI with proposed balancing strategy integrated in PWM technique is simulated using PSCAD/EMTDC and implemented on hardware using CCS. The system is validated using the simulation and experimental results.
- A dc-dc chopper is integrated with ANPCI resulting in CANPCI topology for improved performance. DCM and CCM based current control techniques are proposed to control chopper circuit. The chopper integrated ANPCI with these control techniques is implemented in simulation using PSCAD/EMTDC and on hardware using CCS. The system is validated using the results obtained.
- Development and execution of test cases, acceptance testing strategies for product covering low power and high power electronic components such as DSC, ADC and GPIO interfacing circuits, Gate drivers, IGBTs, etc., and full functional testing.
- A single-phase stiff and weak distribution grid connected solar and wind power generating systems (SWPGS) are proposed using ANPCI to regulate grid power factor. ANPCI is replaced by CANPCI to reduce size of capacitors, effect of 2ω ripple and DC-link capacitor voltage ripples. The controller architecture is carried out for SPGS and PMSG based WPGS, which can also operate at MPPT without solar and wind actuators.
- Average model and its modular structure for ANPCI are proposed considering non-idealities. The model is validated in open-loop and closed-loop using simulation and experimental results. Further, grid-connected and stand-alone SWPGS are modeled with ANPCI and validated using PSCAD/EMTDC simulation results.

Junior Research Fellow (JRF) @ IIT Guwahati

Nov., 2021 – Feb., 2022 and May – Aug., 2022

Design, Development, and demonstration of Solar-PV integrated On-board and Off-board Electric-Rickshaw Charging Infrastructure

- Simulation of PV inverter and testing the solar array simulator.

Assistant Professor @ Gandhi Institute of Technology & Management (GITAM) University, Visakhapatnam

Jun., 2013 – Jul., 2014

Modeling and control of magnetic levitation system

Active power electronic transformer as a power conditioner for non-linear loads

Project Associate @ IIT Hyderabad

Sep., 2011 – Mar., 2012

Design of solar and wind hybrid power system

- System level design of solar and wind hybrid power system and procurement of 10kW wind power system.

M. Tech. project @ Maulana Azad National Institute of Technology (NIT), Bhopal

Jul., 2009 – Aug., 2010

Comparative study of conventional and adaptive control-based vector control of induction motor drive using MATLAB

B. Tech. project @ Raghu Engineering College, Jawaharlal Nehru Technological University (JNTU) Hyderabad

Jul., 2007 – Apr., 2008

Automation in drip irrigation system

- *Hardware:* A regulated power supply, temperature sensor, ADC, LCD, and the Atmel AT89C51 microcontroller.
- *Software:* An assembly language program is developed using KEIL software and deployed into the microcontroller.
- The functional working of the integrated hardware is validated using the experimental results obtained.

Hardware implementation for ± 5V, ± 12V, and ± 15V DC power supply.

IEEE students branch project @ IIIT Hyderabad

May, 2006

Robotic car

- A toy car powered by 9V battery bank is controlled using Atmel®AVR®ATmega32 microcontroller. The controller signals are communicated through IR transceivers to drive the car.

PUBLICATIONS

JOURNALS

1) **Jagath Vallabhai Missula**, Ravindranath Adda, and Praveen Tripathy, "DC-Link Capacitor Voltage Balancing and Ripple Reduction in SRF-dq Controlled Single-Phase ANPCI by Employing External Chopper Circuit," *IEEE Trans. Power Electron. (IF = 5.967) (Under revision for submission).*
2) **Jagath Vallabhai Missula**, Ravindranath Adda, and Praveen Tripathy, "Averaged Modeling and SRF Based Closed-loop Control of Single-phase ANPC Inverter," *IEEE Trans. Power Electron.*, vol. 36, no. 12, pp. 13839–13854, Dec. 2021 *(IF = 5.967).*
3) **Missula Jagath Vallabhai**, Pankaj Swarnkar, and D.M. Deshpande, **"**Comparative Analysis of PI Control and Model Reference Adaptive Control Based Vector Control Strategy for Induction Motor Drive**",** *International Journal of Engineering Research and Applications,* vol. 2, no. 3, pp. 328-335, May–Jun. 2012 *(IF = 5.179).*
4) **Missula Jagath Vallabhai**, Pankaj Swarnkar, and D.M. Deshpande, "PI Control Based Vector Control Strategy for Induction Motor Drive"**,** *International Journal of Electronics Communication and Computer Engineering,* vol.3, no. 2, pp. 2059–2070, Mar. 2012 *(IF = 2.7).*

CONFERENCES

1) Vaishnavvignesh G Iyer, **Jagath Vallabhai Missula**, Cilaveni Satish Chandra, and Ravindranath Adda, "Hardware implementation of PDS-PWM for Five-Level Active Neutral Point Clamped Inverter using LAUNCHXL F28379D", in *Proc. 17th IEEE Energy Convers. Congr. Expo. Asia (ECCE-Asia),* Bengaluru, India, May–2025, pp. 1–6.
2) **Jagath Vallabhai Missula**, Ravindranath Adda, and Praveen Tripathy, "Performance Comparison of Capacitor Voltage Balancing Strategies for Eight-Switch Five-Level ANPC Inverter", in *Proc. 10th National Power Electronics Conf. (NPEC),* Bhubaneswar, India, Dec. 2021, pp. 1–6.
3) **Jagath Vallabhai Missula**, Ravindranath Adda, and Praveen Tripathy, "Reduction of Voltage Ripple for Single-phase Chopper Integrated ANPCI based on Hysteresis Voltage and Chopper Current Control Methods," in *Proc. IEEE Int. Conf. Power Electronics, Drives and Energy Systems (PEDES),* Jaipur, India, Dec. 2020, pp. 1–6.
4) **Jagath Vallabhai Missula**, "Single-phase Five-level Boost Inverter for Stand-alone PV Applications," in *Proc. 46th Annu. Conf. IEEE Ind. Electron. Soc. (IECON),* Marina Bay Sands, Singapore, Oct. 2020, pp. 1481–1486.

5) **Jagath Vallabhai Missula**, Ravindranath Adda, and Praveen Tripathy, "Ripple Reduction in the DC-link Capacitor Voltages of Single-phase ANPC Inverter using External Chopper Circuit," in *Proc. 46th Annu. Conf. IEEE Ind. Electron. Soc. (IECON),* Marina Bay Sands, Singapore, Oct. 2020, pp. 2436−2441.
6) **Jagath Vallabhai Missula**, Ravindranath Adda, and Praveen Tripathy, "A CCM based Average Current Control Technique for Chopper Integrated Single-phase ANPC Inverter to Minimize Voltage Ripple," in *Proc. 12th IEEE Energy Convers. Congr. Expo. (ECCE),* Detroit, MI, USA, Oct. 2020, pp. 2633−2640.
7) **Jagath Vallabhai Missula**, Ravindranath Adda, and Praveen Tripathy, "Average modeling of active neutral point clamped inverter," in *Proc. 11th IEEE Energy Convers. Congr. Expo. (ECCE),* Baltimore, MD, USA, Sep.−Oct. 2019, pp. 3689−3696.
8) **Jagath Vallabhai Missula**, Ravindranath Adda, and Praveen Tripathy, "Pulse Width Modulation and SRF Based Closed-Loop Control of Stand-alone Single-phase 5L-ANPC Inverter", in *Proc. IEEE Int. Conf. Power Electronics, Drives and Energy Systems (PEDES),* Chennai, India, Dec.2018, pp. 1−6.
9) **Jagath Vallabhai Missula**, Ravindranath Adda, and Praveen Tripathy, "Five-Level ANPC Converter for PMSG Based Wind Power System with Grid Power Factor Regulation", in *Proc. 20th National Power Systems Conf. (NPSC),* Tiruchirappalli, India, pp. 1−6, Dec. 2018.
10) **Jagath Vallabhai Missula**, Praveen Tripathy, and Ravindranath Adda, "Wind Energy Conversion System and Storage Technologies for AC Microgrid" in *Proc. Power Conservation Techniques: A Road Map to Green Environment,* Chief Engineer Shillong Zone, pp. 1–8, 27th Oct. 2017.
11) **Missula Jagath Vallabhai**, and Pavan Mehta, "A New Approach for Locking PLL Using AMS Simulation" in *Proc. National Conference on Modeling & Simulation of Electrical Systems (MSES-2013),* 15–16 Feb. 2013.

PARTICIPATION

WORKSHOPS

1) *Go Getter - Behavioral Skill Workshop* organized by Agnito Insights at ePropelled Systems on 17th Nov. 2022.
2) *EV Conclave* organized by E-Mobility lab, IIT Guwahati on 22nd Nov. 2018.
3) *Intra IIT LATEχ* organized by the department of Mathematics, IIT Guwahati on 18th Feb. 2017.
4) *Computational methods for smart grids* organized by the Department of Computer Science & Engineering, IIT Guwahati during 9–13 Mar. 2015.
5) *System Identification and Advances in Control Techniques (SIACT–13)* organized by GITAM University on 28th Dec. 2013.
6) *Patent Awareness* organized by GITAM University on 28th Oct. 2013.
7) *Soft Computing Applications to Solar-Wind Integrated Systems and Energy Auditing* organized by GITAM University on 21st Sep. 2013.

PROFESSIONAL DEVELOPMENT PROGRAMMES

1) *Power Trading Power Exchanges and Merchant Power Plants* organized by Engineering Staff College of INDIA, Hyderabad during 28–30 Jan. 2014.

EXTRA ACTIVITIES

- Reviewer of IEEE Conferences and Journals of PELS and IAS.
- Member of IEEE since 2006.
- Worked as *students' coordinator/counsellor* at GITAM University.
- Worked as an *Academic monitoring committee (AMC)* member at GITAM University.

PERSONAL DETAILS

NAME	:	Jagath Vallabhai Missula
GENDER	:	Male
NATIONALITY	:	Indian
LANGUAGES	:	Telugu, Hindi and English
DATE OF BIRTH	:	14th Aug. 1987
MARITAL STATUS	:	Unmarried